Early Prediction of Diabetes Risk from Routine Clinical Health Parameters Using Machine Learning Models

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Abstract: Diabetes is a major health problem because it leads to heart disease, kidney failure, vision loss, and other long-term complications for human health. Many patients develop the condition after extended periods of uncontrolled blood sugar, high BMI, and irregular blood pressure levels. In Bangladesh, the situation is more difficult due to eating habits, limited routine check-ups, and late recognition of symptoms. As a result, many people come for treatment only after complications have already appeared. Existing studies have approached diabetes prediction by using indicators such as glucose level, BMI, and blood pressure. These works demonstrated that basic clinical measurements could provide an initial view of diabetes risk. Many of the studies were based on small datasets and did not include clinical details such as proper family history, fasting plasma glucose, or HbA1c. Including these factors enables machine learning models to capture risk patterns more accurately and improves their ability to distinguish high-risk individuals. In this research, we aim to develop a forecasting model using machine learning models and a wider range of health parameters along with the current parameters. The selected features include glucose, BMI, blood pressure, fasting plasma glucose, HbA1c, and first-degree family history of diabetes. Considering these values together gives a clearer view of a patient's condition and helps the model learn more meaningful patterns. Machine learning models such as Random Forest, Linear Regression, and Support Vector Machine are used to identify which method gives a more dependable forecast for Bangladeshi patients. These models, combined with the selected clinical features, are expected to show clearer identification of individuals who may develop diabetes in the future. A dependable forecast can guide early clinical progress, and better prevention plans so that patients can manage the risk before the disease becomes serious.

Keywords: Clinical Progress, Early Forecasting, Health Parameter, Machine Learning Model, Patient's condition.

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